

Candidate Name

Centre Number

Candidate

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

PHYSICS**9243/2**

PAPER 2

Friday

1 NOVEMBER 2002

Morning

1 hour 45 minutes

Candidates answer on the question paper.

Additional materials:

Electronic calculator and/or Mathematical tables

Protractor

Ruler (300 mm)

TIME 1 hour 45 minutes**INSTRUCTIONS TO CANDIDATES**

Write your name, Centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

For numerical answers, **all** working should be shown.**INFORMATION FOR CANDIDATES**

The number of marks is given in brackets [] at the end of each question or part question.

The quality of your language will be taken into account in the marking of your answers which contain explanations.

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1	
2	
3	
4	
5	
6	
7	
8	
Quality of language	
TOTAL	

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$$V = \pm \sqrt{2x^2 - 2}$$

Data

speed of light in free space,

$$c = 3.00 \times 10^8 \text{ ms}^{-1}$$

permeability of free space,

$$\mu_0 = 4\pi \times 10^{-7} \text{ Hm}^{-1}$$

permittivity of free space,

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ Fm}^{-1}$$

elementary charge,

$$e = 1.60 \times 10^{-19} \text{ C}$$

the Planck constant,

$$h = 6.63 \times 10^{-34} \text{ Js}$$

unified atomic mass constant,

$$u = 1.66 \times 10^{-27} \text{ kg}$$

rest mass of electron,

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

rest mass of proton,

$$m_p = 1.67 \times 10^{-27} \text{ kg}$$

molar gas constant,

$$R = 8.31 \text{ JK}^{-1} \text{ mol}^{-1}$$

the Avogadro constant,

$$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$$

the Boltzmann constant,

$$k = 1.38 \times 10^{-23} \text{ JK}^{-1}$$

gravitational constant,

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

acceleration of free fall,

$$g = 9.81 \text{ ms}^{-2}$$

Formulae

uniformly accelerated motion,

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

work done on/by a gas,

$$W = p\Delta V$$

gravitational potential,

$$\phi = -\frac{Gm}{r}$$

refractive index,

$$n = \frac{1}{\sin C}$$

resistors in series,

$$R = R_1 + R_2 + \dots$$

resistors in parallel,

$$1/R = 1/R_1 + 1/R_2 + \dots$$

electric potential,

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

capacitors in series,

$$1/C = 1/C_1 + 1/C_2 + \dots$$

capacitors in parallel,

$$C = C_1 + C_2 + \dots$$

energy of charged capacitor,

$$W = \frac{1}{2}QV$$

alternating current/voltage,

$$x = x_0 \sin \omega t$$

hydrostatic pressure,

$$p = \rho gh$$

pressure of an ideal gas,

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

radioactive decay,

$$x = x_0 \exp(-\lambda t)$$

decay constant,

$$\lambda = \frac{0.693}{t_{1/2}}$$

critical density of matter in the Universe,

$$\rho_0 = \frac{3H_0^2}{8\pi G}$$

equation of continuity,

$$Av = \text{constant}$$

Bernoulli equation (simplified),

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

Stokes' law,

$$F = 6\pi\eta r v$$

Reynolds' number,

$$R_e = \frac{\rho v r}{\eta}$$

Drag force in turbulent flow,

$$F = Br^2 \rho v^2$$

1 (a) The following equations may or may not be correct. Use base units to check if they represent the given relationships.

(i) The fundamental frequency f_1 of an open pipe of length L when the speed of sound is v , $f = \frac{v}{2L}$.

(ii) The pressure p at a point in a fluid of density ρ moving with a speed v , $p = \frac{1}{2} \rho v$.

(iii) The angle θ at which an aeroplane moving with a speed v should be banked to describe a path of radius of curvature r , $\tan\theta = \frac{v^2}{rg}$.

[6]

(b) Torque and energy have the same dimensions.

(i) Explain why it would be inappropriate to measure torque in joules.

(ii) Write down an appropriate unit of torque.

[2]

A basketball player shoots a 'free throw' as shown in Fig 2.1 below. He throws the ball at an angle of 55° to the horizontal and gives it an initial velocity u .

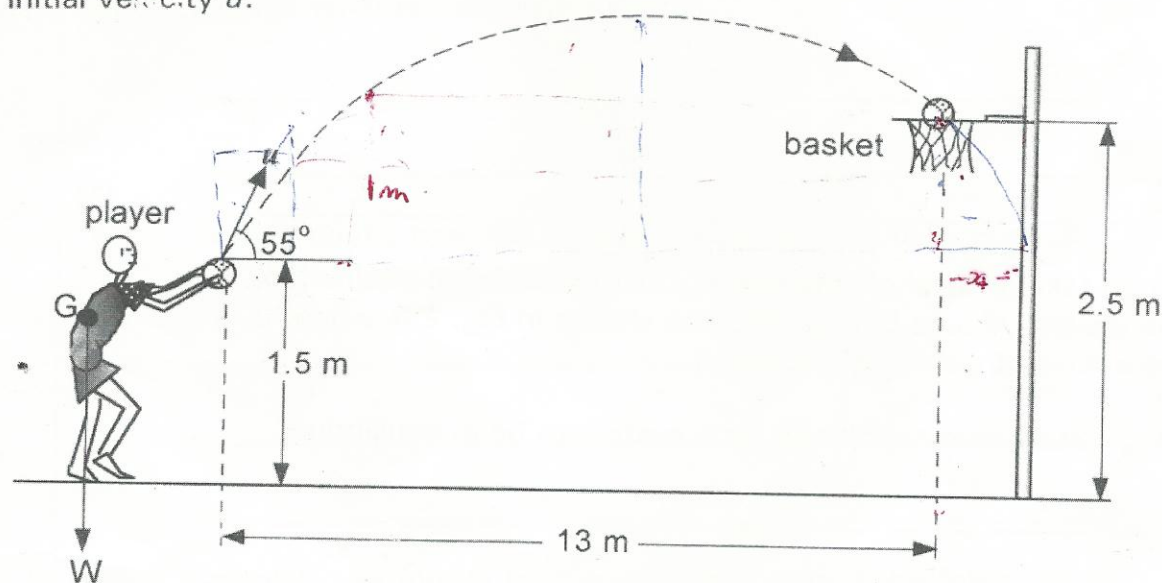


Fig. 2.1

Assume that air resistance is negligible.

(a) What is the component of the ball's initial speed in terms of u ?

(i) horizontally $u \cos 55^\circ$

(ii) vertically $u \sin 55^\circ$

[2]

- (b) Using your answer to a(i), write down an expression for the time t , taken to reach the basket in terms of u .

$t =$ _____ [1]

- (c) Using your answers to (a) and (b), calculate the value of u .

$u =$ _____ [3]

- (d) When taking the shot, the player must be in stable equilibrium. His weight W acts through the line shown in Fig. 2.1 where G is the centre of gravity of his body.

- (i) State **two** conditions for a system to be in equilibrium.

[2]

- * (ii) Suggest **two** ways by which the player can ensure that he remains in equilibrium.

[2]

- 3 A 'torque wrench' is a tool used to tighten nuts and bolts to a precise measurement. One model is shown in Fig. 3.1.

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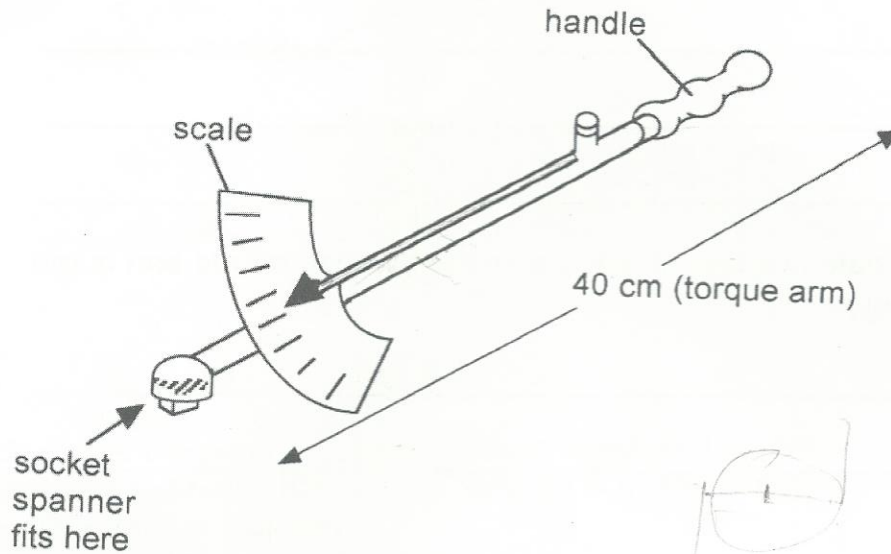


Fig. 3.1

- (a) (i) Explain what is meant by a *torque*.

[2]

- (ii) Explain briefly how the torque wrench is used to tighten a nut by the required amount.

[2]

- (iii) Calculate the torque for the wrench shown if the force applied to the end of the handle is 200 N.

torque = _____

[2]

- (b) (i) Why does a torque wrench have a long handle or torque arm?

[2]

- (ii) State **one** risk of using a wrench whose torque arm is too long.

[1]

- (c) (i) Calculate the distance moved by the end tip of the handle during one complete turn when the force in **a(iii)** is applied.

distance = _____

[2]

- (ii) Hence calculate the energy transferred.

energy transferred = _____

[3]

(d)

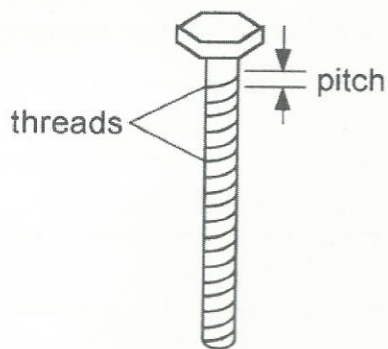


Fig. 3.2

Fig. 3.2 shows the pitch of a screw thread which is 1.5 mm. To stretch the tightened bolt by 1.5 mm, 30% of the energy calculated in 3c(ii) is required.

- (i) Calculated how much force is being used to stretch the bolt.

force = _____ [3]

- (ii) What happens to the other 70% of the energy transferred?

_____ [2]

- 45 (a) (i) What do you understand by *diffraction* of waves?

_____ [2]

- (ii) Demonstrate your appreciation of diffraction by completing Fig. 4.1, showing the wave fronts after passing the following obstacle.

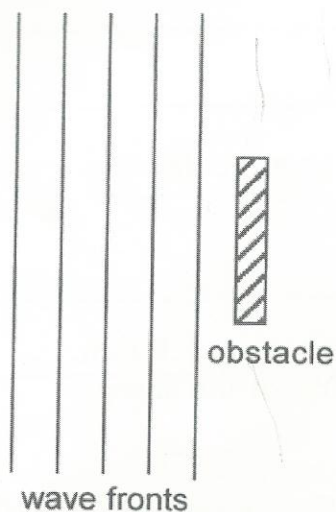
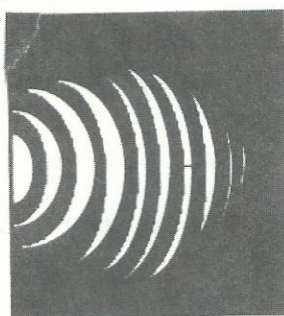


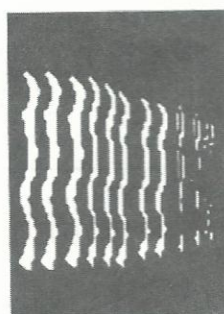
Fig. 4.1

[2]

- (b) Fig. 4.2 shows wave fronts after passing through a narrow and a wide gap positioned on the left-hand end of the pattern.



A



B

Fig. 4.2

- (i) State which one of A or B is the pattern of wave fronts from a narrow gap.

[1]

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✖(c) Why is diffraction of sound waves a common phenomenon but diffraction of light waves is not?

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper appears slightly aged or off-white. There is no handwriting or printed text on the page.

[3]

5

A force acts on a current-carrying conductor placed in a magnetic field. Electromagnetic propulsion devices such as the rail gun (refer to Fig. 5.1) use this principle to propel objects such as a slug.

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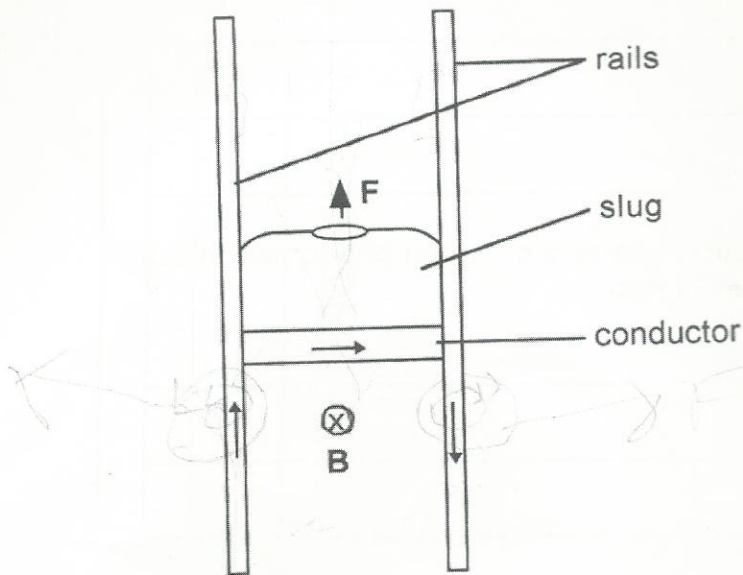


Fig. 5.1

A current that flows through the conducting rails as shown in Fig. 5.1 will give rise to a magnetic field B that is directed into the page. The conductor falls within this field and thus experiences a force F .

- (a) Define the *tesla*, the unit of magnetic flux density.

[1]

- (b) State and explain how the direction of the force F on the conductor is predicted.

[2]

(c) (i) Draw on Fig. 5.1 the magnetic field pattern due to the conducting rails. [2]

(ii) Hence draw arrows showing the forces acting on the rails. [1]

(iii) The forces in (ii) above may present problems. State one possible problem and explain how it can be taken care of.

[2]

(d) State two factors that the force F depends on.

[2]

(e) Suggest one use for such a railgun other than for military purposes.

[1]

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Examiner's
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For two conductors **P** and **Q**, current **I** varies with voltage **V** as shown in Fig. 6.1. below.

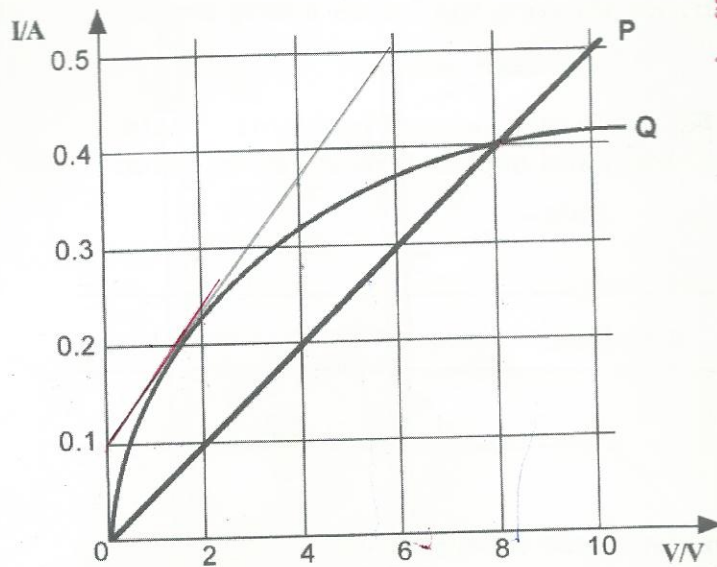


Fig. 6.1

- (a) When **P** and **Q** are connected in series to a voltage supply, a current of 0.2 A flows. Use the graph to estimate the potential difference of the supply (Assume that its internal resistance is negligible).

$$2000 \div 11000 = 0,005$$

$$V = IR$$

$$R = \frac{V}{I}$$

$$\frac{0,2}{4}$$

$$\frac{200}{40}$$

$$0,05$$

$$0,55$$

$$V = IR =$$

potential difference =

[2]

$$R = \frac{V}{I}$$

$$I = \frac{V}{R}$$

$$\frac{1}{0,5} + \frac{1}{0,055}$$

$$\frac{2}{\frac{10}{9}} + \frac{200}{55} = \frac{20}{11}$$

$$0,2 \times 0,55$$

$$110$$

$$\Rightarrow \frac{55 \times 2 + 1000}{55}$$

$$0,11$$

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$$18,2$$

$$\frac{55}{10000} \times \frac{11}{2000}$$

$$\frac{55}{1100}$$

$$\frac{110 + 1000}{55}$$

- (b) The voltage supply in (a) is now connected to a parallel combination of **P** and **Q**. Calculate the current drawn from the supply.

current = _____ [2]

- 7 (a) Give an expression for E , the energy of a photon in terms of f , its frequency, and h the Planck constant.

_____ [1]

$$E = hf$$

- (b) Some of the energy levels of an atom of one substance are shown in Fig. 7.1 below.

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energy in joules $\times 10^{-19}$	energy level n
0.0	∞
-2.56	4
-5.92	3
-8.80	2
-16.64	1

Fig. 7.1

- (i) How much energy is required to raise an electron from $n = 1$ to $n = 4$.

$$E_2 - E_1 = hf$$

energy = _____

[1]

- (ii) Calculate the wavelength of the radiation that is given off by the electron.

wavelength = _____

[2]

- (iii) What part of the electromagnetic spectrum does this radiation belong to?

For
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[1]

- 8 Fig. 8.1 shows the apparatus used to investigate radiant energy.

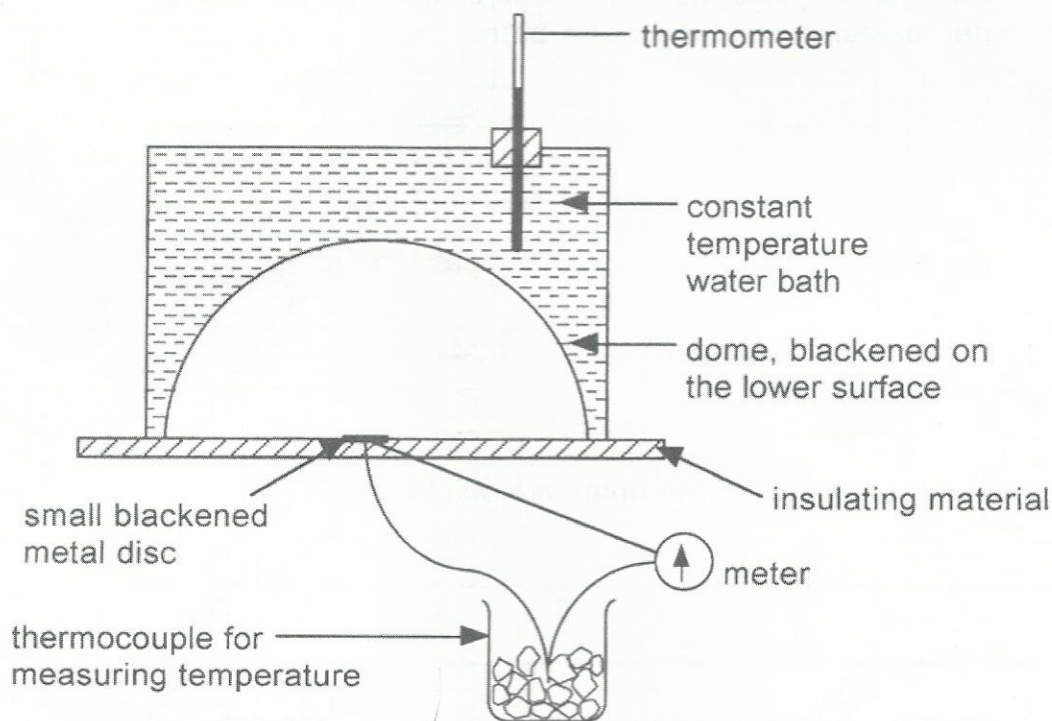


Fig. 8.1

As the blackened metal disc receives radiation from the dome, its temperature, which is measured by the thermocouple rises. The disc is pre-cooled, and then the dome which is at temperature T is put in place over it. Readings are taken for the disc temperature T at a time t as shown on the table below.

t/s	0	15	30	45	60	90	120	150	180
T/K	262	274	283	290	295	303	309	314	317

- (a) Explain why different thermometers used to measure the same temperature may not give the same reading.

[1]

- (b) Suggest two reasons in each case, why

- (i) the liquid-in-glass thermometer is suitable for measuring the temperature of the water bath.

1

2

- (ii) the thermocouple is suitable for measuring temperature T .

1

2

[4]

- (c) Explain why the disc and the dome are blackened.

[2]

- (d) How can you tell from the data that the disc has been pre-cooled.

[1]

- (e) On the following grid, Fig. 8.2, plot a graph of T (y -axis) against t (x -axis).

[4]

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Use

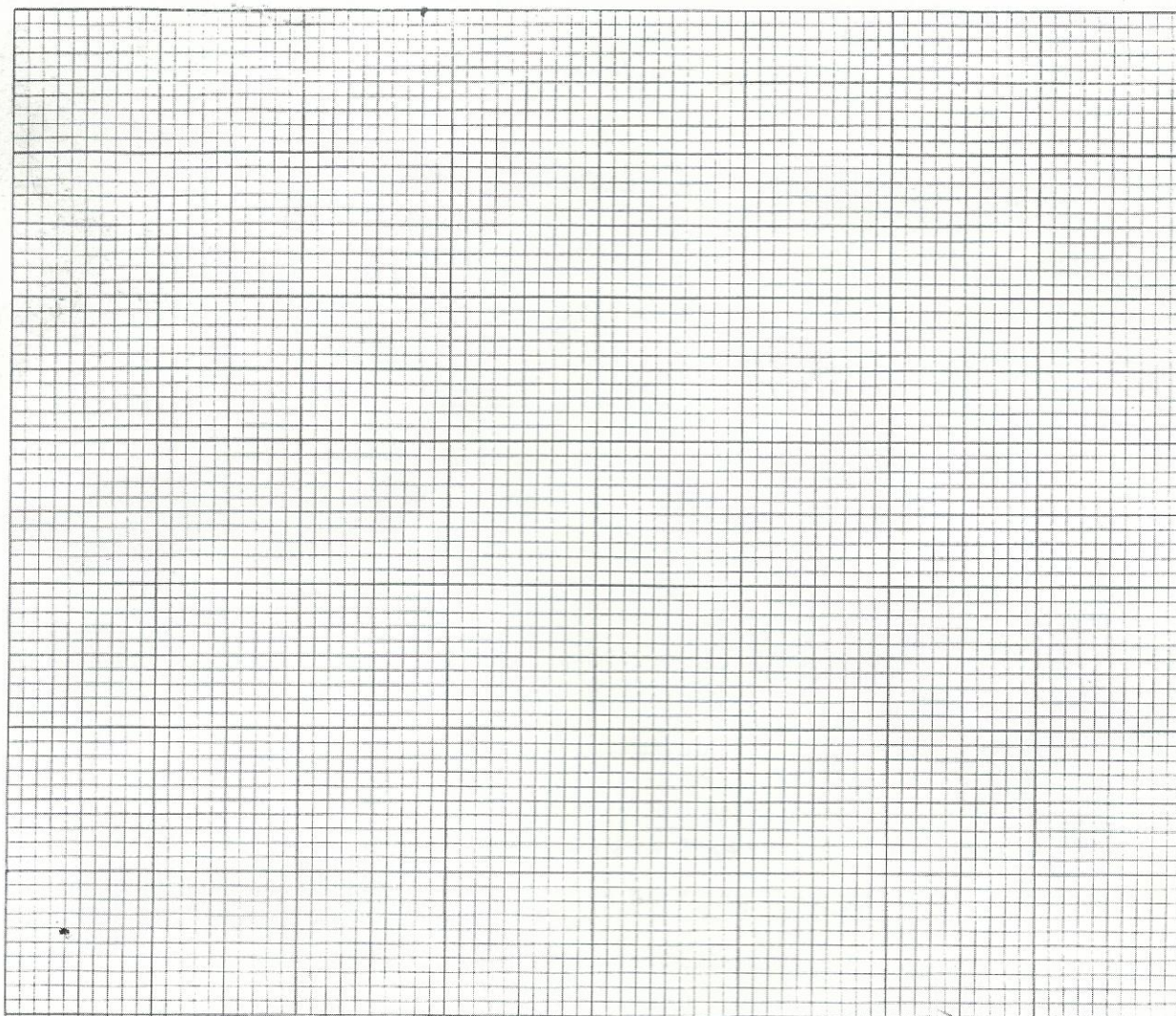


Fig. 8.2

- (f) Use the graph to find the rate of change of temperature R_c at $T = 293$ K.

rate of change of temperature = _____ K/s [3]

(g) Theory suggests that: $\frac{C.R_c}{A} = \delta \left\{ (T_d)^4 - (293)^4 \right\},$

where : C = the heat capacity of the disc = 0.297 JK^{-1}

A = the surface area of the disc = $1.51 \times 10^{-4} \text{ m}^2$

T_d = the temperature of the dome = 373 K

δ = a constant

Determine the value of δ

$\delta =$ _____

[2]

- (h) Similar investigations are carried out with the dome at different values of temperature T_d . In each of these investigations, a graph is drawn and R_c is determined at $T = 293 \text{ K}$.

T_d/K	311	328	345	358	373
R_c/Ks^{-1}	0.056	0.118	0.193	0.251	

Complete the table above by recording your value of R_c for $T_d = 373 \text{ K}$ and then comment on the results.

[1]

Quality of Language

[4]